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Ultrasonic washer and automatic analysis device

Abstract

The present invention provides an ultrasonic washer that radiates ultrasonic waves from a plurality of vibration surfaces, radiates ultrasonic waves at the same phase, effectively generates cavitation around a nozzle through ultrasonic vibration, and has a high washing effect. The ultrasonic washer of the present invention is characterized by having: a washing tank in which is washed a nozzle that suctions a sample or reagent; and an ultrasonic vibrator having a piezoelectric element sandwiched between a front mass and a back mass, wherein a diaphragm between opposing upper and lower plates is formed at a leading end of the front mass installed at a leading end of the ultrasonic vibrator, and the nozzle is ultrasonically washed in a region between the upper plate and the lower plate.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to an automatic analysis device that analyzes the components of a sample by mixing the sample with a reagent, and particularly to an ultrasonic washer and an automatic analysis device that wash nozzles for sample dispensing.

BACKGROUND ART

(2) Since the automatic analysis device uses the same nozzle repeatedly for dispensing samples, it is necessary to perform nozzle washing of washing the leading end of the nozzle with a stream of water before sucking another sample.

(3) However, since an automatic analysis device with high throughput performance dispenses samples at high speed, sufficient time cannot be used for nozzle washing.

(4) For this reason, contaminants derived from the components of the sample may accumulate on the leading end of the nozzle. Accumulation of contaminants on the leading end of the nozzle may cause variations in the amount dispensed and carryover that brings the previous sample into the next sample, and there is a risk that the analysis accuracy may deteriorate.

(5) As a background art in such a technical field, there is WO2017/002740A (PTL 1). PTL 1 describes an ultrasonic washer in which a vibrating unit that magnifies the ultrasonic vibration of a BLT (Bolt Clamped Langevin Transducer) is provided on the side of the inside of a washing tank, and by driving the BLT, cavitation through ultrasonic vibrations is uniformly generated around the nozzles in the washing liquid supplied to the inside of the washing tank and the nozzles are washed effectively (see Abstract).

CITATION LIST

Patent Literature

(6) PTL 1: WO2017/002740A

SUMMARY OF INVENTION

Technical Problem

(7) In general, in an ultrasonic washer, cavitation (a phenomenon in which bubbles are generated and disappear due to a pressure difference occurring in the liquid) becomes stronger in a region where the sound pressure is high, resulting in a high washing effect.

(8) In order to increase the sound pressure, there are methods such as a method of amplifying the vibration of the vibration surface that radiates ultrasonic waves (generates ultrasonic vibration), and a method of irradiating the same region with ultrasonic waves from a plurality of vibration surfaces. In particular, in the case of the method of radiating ultrasonic waves from a plurality of vibration surfaces, it is necessary to radiate the ultrasonic waves at the same phase in order to increase the sound pressure.

(9) PTL 1 describes an ultrasonic washer that uses ultrasonic vibrations of BLT to generate cavitation through ultrasonic vibrations to wash a nozzle.

(10) However, PTL 1 does not describe an ultrasonic washer that radiates ultrasonic waves from a plurality of vibration surfaces, and radiates ultrasonic waves at the same phase in order to increase the sound pressure.

(11) Therefore, the present invention provides an ultrasonic washer and an automatic analysis device that radiate ultrasonic waves from a plurality of vibration surfaces, radiate ultrasonic waves at the same phase, effectively generate cavitation around a nozzle through ultrasonic vibrations, and have a high washing effect.

Solution to Problem

(12) In order to solve the above-described problems, the ultrasonic washer of the present invention is characterized by having a washing tank for washing a nozzle for sucking a sample or a reagent, and an ultrasonic vibrator having a piezoelectric element sandwiched between a front mass and a back mass, in which diaphragms of an upper plate and a lower plate facing each other are formed at a leading end of the front mass installed at a leading end of the ultrasonic vibrator, and the nozzle is

ultrasonically washed and the drive frequency of the ultrasonic vibrator is set between the resonance frequency of the upper plate and the resonance frequency of the lower plate.
(13) Further, an automatic analysis device of the present invention is characterized by having the ultrasonic washer of the present invention.
Advantageous Effects of Invention
(14) According to the present invention, it is possible to provide an ultrasonic washer and an automatic analysis device that radiate ultrasonic waves from a plurality of vibration surfaces, radiate ultrasonic waves at the same phase, effectively generate cavitation around a nozzle through ultrasonic vibration, and have a high washing effect.
(15) Problems, configurations, and effects other than those described above will be clarified by the description of the embodiments below.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is an explanatory diagram illustrating an automatic analysis device **10** having an ultrasonic washer **26** described in a first embodiment.
(2) FIG. **2**A is a top view illustrating the ultrasonic washer **26** described in the first embodiment.
(3) FIG. **2**B is a perspective view illustrating the ultrasonic washer **26** described in the first embodiment.
(4) FIG. **2**C is a cross-sectional view illustrating the ultrasonic washer **26** described in the first embodiment.
(5) FIG. **2**D is an enlarged view of a leading end of an ultrasonic vibrator **201** illustrating the ultrasonic washer **26** described in the first embodiment.
(6) FIGS. **3**A and **3**B are explanatory diagrams illustrating a vibration direction FIG. **3**A and a sound pressure distribution FIG. **3**B when the leading ends (two diaphragms) of a front mass **204** described in the first embodiment vibrate.
(7) FIGS. **3**C and **3**D are explanatory diagrams illustrating the vibration direction FIG. **3**C and the sound pressure distribution FIG. **3**D when the leading ends (two diaphragms) of the front mass **204** described in a comparative example vibrate.
(8) FIGS. **4**A to **4**C are explanatory diagrams illustrating a frequency response of the ultrasonic vibrator **201** and two diaphragms, where FIG. **4**A shows when the resonance frequency of the ultrasonic vibrator **201** is the highest, FIG. **4**B shows when the resonance frequency of the ultrasonic vibrator **201** is the lowest, and FIG. **4**C shows a relationship between the resonance frequency of the ultrasonic vibrator **201** and the resonance frequencies of the two diaphragms, described in the first embodiment.
(9) FIG. **5**A is an explanatory diagram illustrating a leading end shape of the ultrasonic vibrator **201** described in a second embodiment.
(10) FIG. **5**B is an explanatory diagram illustrating a leading end shape of the ultrasonic vibrator **201** described in a third embodiment.
(11) FIG. **5**C is an explanatory diagram illustrating a leading end shape of the ultrasonic vibrator **201** described in a fourth embodiment.
(12) FIG. **5**D is an explanatory diagram illustrating a leading end shape of the ultrasonic vibrator **201** described in a fifth embodiment.
(13) FIG. **5**E is an explanatory diagram illustrating a leading end shape of the ultrasonic vibrator **201** described in a sixth embodiment.
(14) FIG. **5**F is an explanatory diagram illustrating a leading end shape of the ultrasonic vibrator **201** described in a seventh embodiment.
(15) FIGS. **6**A to **6**C are explanatory diagrams illustrating an ultrasonic washer **26** described in an

eighth embodiment, where FIG. **6**A is a perspective view, FIG. **6**B is a cross-sectional view, and FIG. **6**C is a top view.

DESCRIPTION OF EMBODIMENTS

(16) Embodiments of the present invention will be described below with reference to the drawings. Substantially the same or similar configurations are denoted by the same reference numerals, and the description may be omitted if the description is redundant.

First Embodiment

(17) First, the automatic analysis device **10** having the ultrasonic washer **26** described in the first embodiment will be described.

(18) FIG. **1** is an explanatory diagram illustrating the automatic analysis device **10** having the ultrasonic washer **26** described in the first embodiment.

(19) The automatic analysis device **10** analyzes the components of a sample by mixing the sample (specimen) and a reagent. The automatic analysis device **10** includes a reagent disk **12** to be loaded with a plurality of reagent containers **11**, a reaction disk **13** for mixing and reacting reagents and samples, a reagent dispensing mechanism **14** for sucking and discharging reagents, and a sample dispensing mechanism **15** for sucking and discharging the sample.

(20) The reagent dispensing mechanism **14** has a reagent nozzle **21** for dispensing a reagent.

(21) The sample dispensing mechanism **15** has a sample nozzle **22** for dispensing the sample.

(22) In the first embodiment, the automatic analysis device **10** has an ultrasonic washer **26** that ultrasonically washes the leading end of the sample nozzle **22**. The automatic analysis device **10** may have an ultrasonic washer for ultrasonically washing the leading end of the reagent nozzle **21**.

(23) In other words, since the automatic analysis device **10** repeatedly uses the same sample nozzle **22** to dispense the sample, the ultrasonic washer **26** generally performs nozzle washing of washing the leading end of the sample nozzle **22** with a stream of water before sucking another sample.

(24) A sample to be put into the automatic analysis device **10** is placed in a sample container (test tube or blood collection tube) **23**, loaded on a sample rack **24**, and conveyed. A plurality of sample containers **23** are mounted on the sample rack **24**. The sample is blood-derived such as serum or whole blood, urine, or the like.

(25) The sample dispensing mechanism **15** also moves (operates) the sample nozzle **22** to a suction position where the sample is sucked from the sample container **23**, a discharge position where the sample is discharged to a cell **25**, and moves (operates) the sample nozzle **22** to an ultrasonic washing position where the ultrasonic washer **26** that washes the leading end of the sample nozzle **22** is installed and a washing position where a washing tank (not shown) for washing the leading end of the sample nozzle **22** with a stream of water is installed.

(26) The cells **25** are divided into a plurality of cells in the circumferential direction of the reaction disk **13** and installed.

(27) Furthermore, the sample dispensing mechanism **15** adjusts the height of the sample container **23**, the cell **25**, or the ultrasonic washer **26** (washing tank) at the suction position, the discharge position, and the washing position and lowers (operates) the sample nozzle **22**.

(28) In order to perform such an operation, the sample dispensing mechanism **15** rotates the sample nozzle **22** to each position (the suction position, the discharge position, and the washing position) and vertically moves it at each position.

(29) The sample dispensing mechanism **15** is controlled by a drive controller (not shown).

(30) Also, the automatic analysis device **10** has a measuring unit (not shown). Then, the automatic analysis device **10** analyzes (photometrically) the liquid mixture containing the sample and the reagent contained in the cell **25** by the measuring unit. In other words, the automatic analysis device **10** analyzes the concentration of a predetermined component of the sample and analyzes the components of the sample. In addition, the measuring unit has, for example, a light source and a photometer, and the photometer is, for example, an absorption photometer or a scattering photometer.

(31) The ultrasonic washer **26** ultrasonically washes the leading end of the sample nozzle **22** in contact with the sample.
(32) The timing for ultrasonically washing the leading end of the sample nozzle **22** is immediately after the sample dispensing mechanism **15** discharges the sample, after the sample dispensing mechanism **15** finishes dispensing a certain number of times, during the daily first or last maintenance period of the automatic analysis device **10**, or when the automatic analysis device **10** is out of service. The ultrasonic washer **26** is controlled by a drive controller (not shown).
(33) Further, the automatic analysis device **10** may be required to have high throughput performance. In order to dispense a sample at high speed, it is necessary to suck, discharge, and wash the sample in a short time of several seconds during the analysis process. Therefore, it may not be possible to use sufficient time for nozzle washing. In other words, if ultrasonic washing is performed during one cycle of operations (sample suction, discharge, and washing), the throughput performance may be lowered.
(34) Therefore, it is preferable to perform ultrasonic washing according to the type of analysis. In some cases, one type of sample is analyzed by a plurality of analysis devices (for example, low-sensitivity analysis is performed first, followed by high-sensitivity analysis). In such a case, a sequence that improves the overall throughput performance is adopted by performing ultrasonic washing on samples for high-sensitivity analysis and not performing ultrasonic washing when only performing low-sensitivity analysis.
(35) Thus, during the analysis process, ultrasonic washing is performed when a sample analyzed by one measuring unit may be analyzed by another measuring unit, particularly when carryover of sample components from the previous analysis significantly affects the subsequent analysis. This can improve the overall throughput performance. In this case, one washing time is about several seconds.
(36) Also, even at timings other than during analysis process, in order to respond to urgent analysis requests, the automatic analysis device **10** needs to transition to an analysis-ready state in a short period of time and the nozzle washing for a long time of one minute or more is not possible.
(37) As described above, the automatic analysis device **10** requires the ultrasonic washer **26** that efficiently washes the nozzles in a short period of time. In other words, the automatic analysis device **10** requires the ultrasonic washer **26** that effectively generates cavitation around the sample nozzle **22** through ultrasonic vibration and has a high washing effect.
(38) Next, the ultrasonic washer **26** described in the first embodiment will be described.
(39) FIGS. **2**A, **2**B, **2**C, and **2**D are explanatory diagrams illustrating the ultrasonic washer **26** described in the first embodiment, where FIG. **2**A is a top view of the ultrasonic washer **26**, FIG. **2**B is a perspective view of the ultrasonic washer **26**, FIG. **2**C is a cross-sectional view of the ultrasonic washer **26** (A-A cross-section in FIG. **2**A), and FIG. **2**D is an enlarged view of a leading end of the ultrasonic vibrator **201** (front mass **204**).
(40) The ultrasonic washer **26** includes an ultrasonic vibrator **201** and a base **203** integrally formed with a washing tank **202** for washing the leading end of the sample nozzle **22** for sucking a sample.
(41) The ultrasonic vibrator **201** includes a front mass (washing head) **204**, a plurality of piezoelectric elements **205** (for example, piezoelectric ceramics), a plurality of electrodes **206**, and a back mass **207**.
(42) Further, a flange **209** for fixing the ultrasonic vibrator **201** to the base **203** is installed on the front mass **204**. By inserting the flange **209** on which the ultrasonic vibrator **201** is installed into the slit provided in the base **203**, the position of the ultrasonic vibrator **201** in the X and Y directions is fixed.
(43) A hole (washing hole) **210** into which the leading end of the sample nozzle **22** is inserted is formed in the leading end of the front mass **204** (the upper plate **221** and the lower plate **222**, which are the diaphragms to be described later). Then, the sample nozzle **22** is lowered toward the washing hole **210** and immersed in the washing liquid stored inside the washing tank **202**.

(44) In the first embodiment, the ultrasonic vibrator **201** is installed at an angle close to horizontal (or horizontally) with respect to the liquid surface of the washing liquid filled in the washing tank **202** (the liquid surface of the washing tank **202**) so that the angle between the ultrasonic wave irradiation surface of the diaphragm and the liquid surface of the washing tank **202** is nearly parallel (or parallel).
(45) The ultrasonic vibrator **201** is fixed by having the piezoelectric element **205** and the electrode **206** sandwiched between the front mass **204** and the back mass **207** and being bolt-clamped with a bolt **208** in the same manner as a general BLT (Bolt Clamped Langevin Transducer). See FIG. **2**C.
(46) Further, a channel **211** is formed in the washing tank **202**. The channel **211** is connected to a channel communicating with a washing liquid tank (not shown) and a syringe pump (not shown) installed in the automatic analysis device **10**. As a result, the washing liquid is supplied to the washing tank **202** and the washing liquid is recovered from the washing tank **202**. See FIG. **2**C.
(47) It is also possible to form a groove for receiving the washing liquid outside the washing tank **202**, to supply the washing liquid to the washing tank **202**, to overflow the washing liquid, and to recover the washing liquid.
(48) Thus, at the leading end of the front mass **204** installed at the leading end of the ultrasonic vibrator **201** described in the first embodiment, two diaphragms of the upper plate **221** and the lower plate **222**, which vibrate in opposite phases (driven at frequencies that vibrate in opposite phases), are formed to face each other. See FIG. **2**D.
(49) The liquid level of the washing liquid stored inside the washing tank **202** is preferably at least as high as the upper plate **221** or higher. That is, the space between the upper plate **221** and the lower plate **222** needs to be filled with the washing liquid.
(50) The internal size (size and depth) of the washing tank **202** only needs to be that the washing tank **202** does not come into contact with the front mass **204**, and that the upper plate **221** and the lower plate **222**, which are two diaphragms formed at the leading end of the front mass **204**, can be immersed in the washing liquid.
(51) Furthermore, in the first embodiment, the shapes of the upper plate **221** and the lower plate **222**, which are the two diaphragms, are changed.
(52) In particular, in the first embodiment, a portion (partially thick portion (shape): thick portion) having a different thickness is formed in a part (root portion) of the upper plate **221**, and the difference in mass between the upper plate **221** and the lower plate **222** changes the rigidity of the upper plate **221** and the lower plate **222**.
(53) The thick portion is formed in either the upper plate **221** or the lower plate **222**, and is formed in the root portion or leading end portion of the diaphragm.
(54) That is, in the first embodiment, the weights of the upper plate **221** and the lower plate **222** are changed in order to change the rigidity of the upper plate **221** and the lower plate **222**. In particular, in the first embodiment, the weight of the lower plate **222** is made lighter than the weight of the upper plate **221**. Thereby, the upper plate **221** and the lower plate **222** can be vibrated in opposite phases. The weight of the lower plate **222** may be made heavier than the weight of the upper plate **221**.
(55) The upper plate **221** and the lower plate **222** are, for example, 4 mm wide, 11 mm long in the longitudinal direction, 1 mm thick, and 2.5 mm thick at the root portion of the upper plate **221**. The distance between the upper plate **221** and the lower plate **222** is 2 mm, the diameter of the washing hole **210** is 3 mm, and the longitudinal length of the root portion between the upper plate **221** and the lower plate **222** is 3 mm.
(56) Also, the closer the distance between the upper plate **221** and the lower plate **222** is, the more effective the sound pressure is. Therefore, it is preferable that the distance between the upper plate **221** and the lower plate **222** is about 10 mm at maximum. The range of contact between the sample nozzle **22** and the sample is about several millimeters, and the sample nozzle **22** can be sufficiently washed.

(57) Further, the sample nozzle **22** can be vertically moved by the sample dispensing mechanism **15**. Therefore, the sample nozzle **22** can be moved up and down with respect to the region between the upper plate **221** and the lower plate **222** to wash a wide range. For example, first, several millimeters of the leading end portion of the sample nozzle **22** are washed, the sample nozzle **22** is lowered by several millimeters, the next several millimeters are washed, the sample nozzle **22** is lowered by several millimeters again, and then the next several millimeters are washed. By repeating this operation, a wide range can be washed. Alternatively, the sample nozzle **22** may be inserted deeply first, and the washing may be performed while the sample nozzle **22** is raised. Washing can be performed regardless of the moving direction and moving distance of the sample nozzle **22**.
(58) Thus, in the automatic analysis device **10** described in the first embodiment, when the sample nozzle **22** is ultrasonically washed in the region between the upper plate **221** and the lower plate **222**, ultrasonic washing is performed while changing the height of the sample nozzle **22**.
(59) In the first embodiment, in order to change the mass of the upper plate **221** and the mass of the lower plate **222** while using the same material (density) for the upper plate **221** and the lower plate **222**, the length, thickness, and amount (forming protrusions and thick portions) are changed. Moreover, in order to change the mass of the upper plate **221** and the mass of the lower plate **222**, the upper plate **221** and the lower plate **222** may be made of different materials (density) with the same shape.
(60) In addition, the length, thickness, and amount (forming protrusions and thick portions) may be combined and changed. For example, the thickness and length of the upper plate **221** and the lower plate **222** may be changed. Also, in a shape having different thicknesses and lengths, the widths of the upper plate **221** and the lower plate **222** may be widened. Also, the washing hole **210** may have a shape in which the washing hole **210** is cut open to the leading end instead of being a hole.
(61) In the first embodiment, by changing the rigidity of the upper plate **221** and the lower plate **222**, the upper plate **221** and the lower plate **222** can be vibrated in opposite phases.
(62) By vibrating the upper plate **221** and the lower plate **222** in opposite phases, it is possible to generate a region where the sound pressure is concentrated (amplified) in the region (hollow portion) between the upper plate **221** and the lower plate **222**. That is, by vibrating the upper plate **221** and the lower plate **222** in opposite phases, it is possible to radiate ultrasonic waves in the same phase to the hollow portion, and by radiating the ultrasonic waves in the same phase, it is possible to generate a region where the sound pressure is concentrated.
(63) As described above, according to the first embodiment, by radiating ultrasonic waves in the same phase from a plurality of vibration surfaces (two planes), it is possible to generate a region where sound pressure is concentrated (a region where sound pressure is increased).
(64) A good washing effect can be obtained by inserting the leading end of the sample nozzle **22** into the region where the sound pressure is concentrated.
(65) As described above, the ultrasonic washer **26** described in the first embodiment has the ultrasonic vibrator **201**, and the washing head (front mass **204**) is installed at the leading end of the ultrasonic vibrator **201**, and at the leading end (end portion) of the washing head (the front mass **204**), there are the upper plate **221** and the lower plate **222** which serve as diaphragms and have different rigidity. As a material for the diaphragm, for example, a chemical-resistant SUS material that can be used with an alkaline detergent is used.
(66) By changing the length, thickness, and mass of the upper plate **221** and the lower plate **222**, the rigidity of the upper plate **221** and the lower plate **222** is changed.
(67) As described above, according to the first embodiment, the automatic analysis device **10** having the ultrasonic washer **26** having a high washing effect can be provided in which cavitation through ultrasonic vibration is effectively generated around the sample nozzle **22**, and the washing effect of effectively washing contaminants adhering to the leading end of the sample nozzle **22** is achieved.

(68) Here, the vibration direction (a) and the sound pressure distribution (b) when the leading end (two diaphragms) of the front mass **204** vibrates will be described.
(69) FIG. **3**A is an explanatory diagram illustrating the vibration direction (a) and the sound pressure distribution (b) when the leading end (two diaphragms) of the front mass **204** described in the first embodiment vibrates.
(70) FIG. **3**B is an explanatory diagram illustrating the vibration direction (a) and the sound pressure distribution (b) when the leading end (two diaphragms) of the front mass **204** described in the comparative example vibrates.
(71) When the front mass **204** is vibrated, the upper plate **221** and the lower plate **222**, which are two diaphragms formed at the leading end of the front mass **204**, vibrate in opposite phases or in the same phase.
(72) FIG. **3**A(a) shows the deformed shapes of the upper plate **221** and the lower plate **222** when vibrating in opposite phases, and FIG. **3**B(a) shows the deformed shapes of the upper plate **221** and the lower plate **222** when vibrating in the same phase.
(73) As shown in FIG. **3**A(a), in opposite phases, the upper plate **221** and the lower plate **222** vibrate in opposite directions. Therefore, when the upper plate **221** deforms downward (−Z direction), the lower plate **222** deforms upward (+Z direction), and when the upper plate **221** deforms upward (+Z direction), the lower plate **222** deforms downward (−Z direction).
(74) On the other hand, as shown in FIG. **3**B(a), in the same phase, the upper plate **221** and the lower plate **222** vibrate in the same direction. Therefore, when the upper plate **221** deforms upward (+Z direction), the lower plate **222** also deforms upward (+Z direction), and when the upper plate **221** deforms downward (−Z direction), the lower plate **222** also deforms downward (−Z direction).
(75) FIG. **3**A(b) shows the phase (sound pressure distribution) of ultrasonic waves generated when the upper plate **221** and the lower plate **222** vibrate in opposite phases, and FIG. **3**B(b) shows the phase (sound pressure distribution) of ultrasonic waves generated when the upper plate **221** and the lower plate **222** vibrate in the same phase.
(76) As shown in FIG. **3**A(b), when the upper plate **221** and the lower plate **222** are driven at frequencies that vibrate in opposite phases, the vibrations of ultrasonic waves generated in the region (hollow portion) between the upper plate **221** and the lower plate **222** have the same phase and act in the direction of increasing the sound pressure.
(77) On the other hand, as shown in FIG. **3**B(b), when the upper plate **221** and the lower plate **222** are driven at frequencies that vibrate in the same phase, the vibrations of ultrasonic waves generated in the region (hollow portion) between the upper plate **221** and the lower plate **222** have opposite phases and act in the direction of weakening the sound pressure (the direction in which the ultrasonic waves cancel each other (the sound pressure is canceled)).
(78) As described above, according to the first embodiment, the upper plate **221** and the lower plate **222** are vibrated in opposite phases, and the hollow portion is irradiated with ultrasonic waves in the same phase, so that a region where the sound pressure is concentrated (a region where the sound pressure increases (strengthens)) can be generated in the hollow portion, and cavitation through ultrasonic vibration can be effectively generated around the sample nozzle **22**.
(79) In the first embodiment, the diaphragm uses the secondary vibration mode (displacement near the center is large) in order to increase the sound pressure in the region where the washing hole **210** is located near the center of the diaphragm.
(80) The diaphragm can also use a primary vibration mode (displacement on the leading end side is large).
(81) In this case, since a region where the sound pressure is increased is generated on the leading end side where the displacement is large, it is preferable to form the washing hole **210** into which the sample nozzle **22** is inserted on the leading end side of the diaphragm.
(82) However, in order to excite the primary vibration mode in the frequency band of 20 kHz or higher used in ultrasonic washing, the rigidity needs to be increased by shortening the length of the

diaphragm or increasing the thickness of the diaphragm. In this case, if the length of the diaphragm is shortened, the region irradiated with the ultrasonic waves becomes narrower, and if the thickness of the diaphragm is increased, the depth of the washing tank **202** increases.
(83) Therefore, it is preferable that the vibrations of the upper plate **221** and the lower plate **222** become in a state close to opposite phases by forming the washing hole **210** near the center of the diaphragm and using the secondary vibration mode in which the displacement near the center of the diaphragm is maximum.
(84) That is, it is preferable that the upper plate **221** and the lower plate **222** have the maximum amplitude near the position where the sample nozzle **22** is inserted (the position where the washing hole **210** is formed).
(85) Next, the frequency response of the ultrasonic vibrator **201** and the two diaphragms (the upper plate **221** and the lower plate **222**) will be described.
(86) FIG. **4** is an explanatory diagram illustrating the frequency response of the ultrasonic vibrator **201** and the two diaphragms, where (a) shows when the resonance frequency of the ultrasonic vibrator **201** is the highest, (b) shows when the resonance frequency of the ultrasonic vibrator **201** is the lowest, and (c) shows the relationship between the resonance frequency of the ultrasonic vibrator **201** described in the first embodiment and the resonance frequencies of the two diaphragms.
(87) The ultrasonic vibrator **201** has a resonance frequency f**1** (**401**) at which the piezoelectric element **205** expands and contracts greatly in the X direction. The resonance frequency f**1** (**401**) is determined by the length of the ultrasonic vibrator **201** in the X direction and the speed of sound of the front mass **204**, the back mass **207**, and the piezoelectric element **205**.
(88) The resonance frequency f**2** (**402**) of the upper plate **221** and the resonance frequency f**3** (**403**) of the lower plate **222**, at which the upper plate **221** and the lower plate **222** are greatly deformed in the Z direction as shown in FIG. **3**, are determined by the rigidity such as the thickness and length of the upper plate **221** and the rigidity such as the thickness and length of the lower plate **222**.
(89) The resonance frequency f**1** (**401**) at which the ultrasonic vibrator **201** expands and contracts changes only about several kHz even when the shape of the upper plate **221** and the shape of the lower plate **222** are changed.
(90) The setting patterns of the resonance frequencies of f**1**, f**2**, and f**3** include the order of f**2**, f**3**, and f**1** as shown in FIG. **4**(*a*), the order of f**1**, f**2**, and f**3** as shown in FIG. **4**(*b*), and the order of f**2**, f**1**, and f**3** as shown in FIG. **4**(*c*). The order of f**2** and f**3** may be reversed.
(91) That is, FIG. **4**(*a*) shows f**2** and f**3** lower than f**1**, FIG. **4**(*b*) shows f**2** and f**3** higher than f**1**, and FIG. **4**(*c*) shows f**1** between f**2** and f**3**.
(92) In such a setting pattern of the resonance frequency, when the drive frequency of the ultrasonic vibrator **201** (frequency for driving the ultrasonic vibrator **201**) is around f**1**, the condition for vibrating the upper plate **221** and the lower plate **222** in opposite phases is only when f**1** (drive frequency) is between f**2** and f**3**, as shown in FIG. **4**(*c*). That is, the drive frequency (resonance frequency f**1**) is between a certain resonance frequency (f**2** or f**3**) at which the deformation of the diaphragm becomes large and another resonance frequency (f**3** or f**2**) at which the deformation of the diaphragm becomes large.
(93) The difference in rigidity between the upper plate **221** and the upper plate **222** can be designed by adjusting the difference in length, the difference in thickness, and partial changes in thickness (protrusions and thick portions). That is, the frequency difference between f**2** (**402**) and f**3** (**403**) can be set.
(94) Thus, by setting the resonance frequency f**2** of the upper plate **221** and the resonance frequency f**3** of the lower plate **222** around the resonance frequency f**1** at which the ultrasonic vibrator **201** expands and contracts, and setting the drive frequency to the resonance frequency f**1**, the vibration phases of the upper plate **221** and the lower plate **222** are opposite to each other. Thus, ultrasonic waves with the same phase are generated between the two diaphragms that vibrate in

opposite phases to increase the sound pressure and obtain a good washing effect.
(95) Furthermore, in order to amplify the amplitude of the upper plate **221** and the lower plate **222** with opposite-phase vibration, f**2** and f**3** are preferably close to f**1**, and f**2** and f**3** are each preferably within about 5 kHz from f**1**. For example, if f**1** is 40 kHz, f**2** is set to 35 kHz and f**3** is set to 45 kHz. f**2** and f**3** may be reversed.
(96) That is, it is preferable to set the drive frequency (near f**1**) between f**2** and f**3**, and set f**2** and f**3** close to f**1** (f**2** and f**3** are each within about 5 kHz from f**1**). Thus, the frequency difference between f**2** and f**3** is preferably within 10 kHz.
(97) By setting the drive frequency in this manner, the upper plate **221** and the lower plate **222** can be vibrated in opposite phases, and the amplitudes of both the upper plate **221** and the lower plate **222** can be amplified.
(98) Even when the drive frequency is set to a frequency band other than around f**1**, there are conditions under which the upper plate **221** and the lower plate **222** vibrate in opposite phases. However, in this case, the amplitude of the upper plate **221** or the lower plate **222** may decrease compared to the case where the drive frequency is in the frequency band around f**1**. Therefore, it is preferable to set the drive frequency to a frequency band near f**1**.
(99) Even when the drive frequency is set to a frequency band other than around f**1**, as long as the upper plate **221** and the lower plate **222** vibrate in opposite phases, it is possible to generate a region where sound pressure is concentrated.
(100) As described above, the ultrasonic vibrator **201** has three characteristic resonance frequencies. One is the frequency at which the amplitude is amplified by axial expansion and contraction of the ultrasonic vibrator **201** (referred to as the vibrator expansion and contraction frequency) and the remaining two are frequencies at which the amplitude in the direction perpendicular to the surfaces of the upper plate **221** and the lower plate **222** is amplified (referred to as the upper plate frequency and the lower plate frequency).
(101) According to the first embodiment, the ultrasonic vibrator **201** sets the vibrator expansion and contraction frequency between the upper plate frequency and the lower plate frequency, and drives the ultrasonic vibrator **201** at the vibrator expansion and contraction frequency during nozzle washing, whereby the two diaphragms can be vibrated in opposite phases.
(102) The automatic analysis device **10** also includes a drive controller (not shown) for driving the ultrasonic washer **26**. The drive controller inputs a sinusoidal voltage of, for example, 20 kHz or more to the ultrasonic vibrator **201** to ultrasonically vibrate the diaphragms (the upper plate **221** and the lower plate **222**).
(103) Thus, the vibration phases of the upper plate **221** and the lower plate **222** are opposite phases that are reversed by about 180 degrees, and the ultrasonic washer **26** generates cavitation in a region between the two diaphragms to wash the sample nozzle **22**.
(104) The drive control unit has an automatic follow-up function that automatically follows the resonance frequency f**1** of the ultrasonic vibrator **201**, and sets the drive frequency to the resonance frequency f**1**. In general, the resonance frequency of the ultrasonic vibrator **201** has a high Q value (sharpness), and if the drive frequency deviates from the resonance frequency f**1**, the amplitude of the ultrasonic vibrator **201** may decrease. Therefore, it is preferable to set the drive frequency to the resonance frequency f**1** with the lowest impedance (high drive current). In other words, the drive frequency is preferably the resonance frequency at which the displacement when the ultrasonic vibrator **201** expands and contracts is maximized.
(105) Moreover, in the first embodiment, two diaphragms are used which are easy to design and manufacture. Three or more diaphragms can also be used. The sample nozzle **22** can be washed by vibrating the opposing diaphragms in opposite phases and inserting the sample nozzle **22** into the region between the diaphragms.
(106) In addition, in the automatic analysis device **10**, the installation position of the ultrasonic washer **26** may be any place in the automatic analysis device **10**, and the ultrasonic washer **26** may

be installed only when moving to the washing position or when washing the nozzle.
(107) Further, in the first embodiment, the nozzle to be washed is the sample nozzle **22**. The nozzle to be washed is not limited to the sample nozzle **22**, and the reagent nozzle **21** may also be used.
Second Embodiment
(108) Next, the leading end shape of the ultrasonic vibrator **201** described in a second embodiment will be described.
(109) FIG. **5**A is an explanatory diagram illustrating the leading end shape of the ultrasonic vibrator **201** described in the second embodiment.
(110) Two diaphragms of the upper plate **221** and the lower plate **222** formed at the leading end of the front mass **204** installed at the leading end of the ultrasonic vibrator **201** described in the second embodiment have a shape different from that of the two diaphragms described in the first embodiment. Other parts are the same as in the first embodiment.
(111) That is, in the second embodiment, the thicknesses of the upper plate **221** and the lower plate **222** are changed in order to change the rigidity of the upper plate **221** and the lower plate **222**. In particular, in the second embodiment, the thickness of the lower plate **222** is made thicker than the thickness of the upper plate **221**. Thereby, the upper plate **221** and the lower plate **222** can be vibrated in opposite phases. The thickness of the lower plate **222** may be thinner than the thickness of the upper plate **221**.
Third Embodiment
(112) Next, the leading end shape of the ultrasonic vibrator **201** described in a third embodiment will be described.
(113) FIG. **5**B is an explanatory diagram illustrating the leading end shape of the ultrasonic vibrator **201** described in the third embodiment.
(114) Two diaphragms of the upper plate **221** and the lower plate **222** formed at the leading end of the front mass **204** installed at the leading end of the ultrasonic vibrator **201** described in the third embodiment also have a shape different from that of the two diaphragms described in the first embodiment. Other parts are the same as in the first embodiment.
(115) That is, in the third embodiment, the lengths of the upper plate **221** and the lower plate **222** are changed in order to change the rigidity of the upper plate **221** and the lower plate **222**. In particular, in the third embodiment, the length of the lower plate **222** is made shorter than the length of the upper plate **221**. Thereby, the upper plate **221** and the lower plate **222** can be vibrated in opposite phases. The length of the lower plate **222** may be longer than the length of the upper plate **221**.
Fourth Embodiment
(116) Next, the leading end shape of the ultrasonic vibrator **201** described in a fourth embodiment will be described.
(117) FIG. **5**C is an explanatory diagram illustrating the leading end shape of the ultrasonic vibrator **201** described in the fourth embodiment.
(118) Two diaphragms of the upper plate **221** and the lower plate **222** formed at the leading end of the front mass **204** installed at the leading end of the ultrasonic vibrator **201** described in the fourth embodiment also have a shape different from that of the two diaphragms described the first embodiment. Other parts are the same as in the first embodiment.
(119) That is, in the fourth embodiment, the weights of the upper plate **221** and the lower plate **222** are changed in order to change the rigidity of the upper plate **221** and the lower plate **222**. In particular, in the fourth embodiment, the weight of the lower plate **222** is made lighter than the weight of the upper plate **221**. Thereby, the upper plate **221** and the lower plate **222** can be vibrated in opposite phases. The weight of the lower plate **222** may be made heavier than the weight of the upper plate **221**.
(120) In particular, in the fourth embodiment, a portion with a different thickness (portion with a thicker thickness: protrusion) is formed in a part (leading end portion) of the upper plate **221** and

the rigidity of the upper plate **221** and the lower plate **222** is changed.

Fifth Embodiment

(121) Next, the leading end shape of the ultrasonic vibrator **201** described in a fifth embodiment will be described.

(122) FIG. **5**D is an explanatory diagram illustrating the leading end shape of the ultrasonic vibrator **201** described in the fifth embodiment.

(123) The two diaphragms of the upper plate **221** and the lower plate **222** formed at the leading end of the front mass **204** installed at the leading end of the ultrasonic vibrator **201** described in the fifth embodiment also have a shape different from that of the two diaphragms described in the first embodiment. Other parts are the same as in the first embodiment.

(124) That is, in the fifth embodiment, the weights of the upper plate **221** and the lower plate **222** are changed in order to change the rigidity of the upper plate **221** and the lower plate **222**. In particular, in the fifth embodiment, the weight of the lower plate **222** is made heavier than the weight of the upper plate **221**. Thereby, the upper plate **221** and the lower plate **222** can be vibrated in opposite phases. The weight of the lower plate **222** may be lighter than the weight of the upper plate **221**.

(125) In particular, in the fifth embodiment, a portion with a different thickness (portion with a thicker thickness: thick portion) is formed in a part (root portion) of the lower plate **222**, and the rigidity of the upper plate **221** and the lower plate **222** is changed.

Sixth Embodiment

(126) Next, the leading end shape of the ultrasonic vibrator **201** described in a sixth embodiment will be described.

(127) FIG. **5**E is an explanatory diagram illustrating the leading end shape of the ultrasonic vibrator **201** described in the sixth embodiment.

(128) The two diaphragms of the upper plate **221** and the lower plate **222** formed at the leading end of the front mass **204** installed at the leading end of the ultrasonic vibrator **201** described in the sixth embodiment also have a shape different from that of the two diaphragms described in the first embodiment. Other parts are the same as in the first embodiment.

(129) That is, in the sixth embodiment, in order to change the rigidity of the upper plate **221** and the lower plate **222**, the weights of the upper plate **221** and the lower plate **222** are changed, and the widths of the upper plate **221** and the lower plate **222** are widened compared to the first embodiment. In particular, in the sixth embodiment, the weight of the upper plate **221** is made heavier than the weight of the lower plate **222** as in the first embodiment. Thereby, the upper plate **221** and the lower plate **222** can be vibrated in opposite phases. The weight of the upper plate **221** may be lighter than the weight of the lower plate **222**.

(130) In particular, in the sixth embodiment, the widths of the upper plate **221** and the lower plate **222** are widened, and a portion with a different thickness (portion with a thicker thickness: thick portion) is formed in a part (root portion) of the upper plate **221**, and the rigidity of the upper plate **221** and the lower plate **222** is changed.

(131) The width of the upper plate **221** and the lower plate **222** is, for example, 6 mm (4 mm in the first embodiment). Other parts are the same as in the first embodiment.

(132) Thus, in the sixth embodiment, the widths of the diaphragms (the widths of the upper plate **221** and the lower plate **222**) are preferably wider than the width of the base of the front mass **204** on which the diaphragms are installed.

(133) In addition, the sound pressure concentration effect due to the opposite phase is preferable because the area of ultrasonic wave irradiation increases as the area between the upper plate **221** and the lower plate **222** increases.

Seventh Embodiment

(134) Next, the leading end shape of the ultrasonic vibrator **201** described in a seventh embodiment will be described.

(135) FIG. **5**F is an explanatory diagram illustrating the leading end shape of the ultrasonic vibrator **201** described in the seventh embodiment.
(136) Two diaphragms of the upper plate **221** and the lower plate **222** formed at the leading end of the front mass **204** installed at the leading end of the ultrasonic vibrator **201** described in the seventh embodiment also have a shape different from that of the two diaphragms described in the first embodiment. Other parts are the same as in the first embodiment.
(137) That is, in the seventh embodiment, in order to change the rigidity of the upper plate **221** and the lower plate **222**, the thicknesses of the upper plate **221** and the lower plate **222** are changed, and the washing hole **210** into which the sample nozzle **22** is inserted is cut open to the leading end. In particular, in the seventh embodiment, the thickness of the lower plate **222** is made thicker than the thickness of the upper plate **221** as in the second embodiment. Thereby, the upper plate **221** and the lower plate **222** can be vibrated in opposite phases. The thickness of the lower plate **222** may be thinner than the thickness of the upper plate **221**.
Eighth Embodiment
(138) Next, the ultrasonic washer **26** described in an eighth embodiment will be described.
(139) FIG. **6** is an explanatory diagram illustrating the ultrasonic washer **26** described in the eighth embodiment, where (a) is a perspective view of the ultrasonic washer **26**, (b) is a cross-sectional view of the ultrasonic washer **26** (B-B cross section of (c)), and (c) is a top view of the ultrasonic washer **26**.
(140) The ultrasonic washer **26** described in the eighth embodiment differs from the ultrasonic washer **26** described in the first embodiment in the shape of the front mass **204** and the installation direction of the front mass **204**.
(141) In the first embodiment, the ultrasonic vibrator **201** is installed at an angle close to horizontal with respect to the liquid surface of the washing tank **202**. On the other hand, in the eighth embodiment, the ultrasonic vibrator **201** is installed at an angle close to vertical (or vertical) with respect to the liquid surface of the washing tank **202**.
(142) That is, in the eighth embodiment, the upper plate **221** and the lower plate **222**, which are two diaphragms formed at the leading end of the front mass **204**, are installed so as to be orthogonal to the main body of the front mass **204**.
(143) In addition, in the eighth embodiment, two diaphragms are installed on the left and right (at two locations) with respect to the main body of the front mass **204**. Two diaphragms installed at two locations are immersed in the washing liquid stored inside the washing tank **202**.
(144) The ultrasonic vibrator **201** is fixed by inserting the flange **209** into a slit provided in the base (not shown), as in the first embodiment.
(145) The front mass **204** has two upper plates **221** and two lower plates **222**, for a total of four diaphragms. Then, an upper plate **221***a* and a lower plate **222***a*, and an upper plate **221***b* and a lower plate **222***b* form pairs, respectively, and each pair forms a region for increasing the sound pressure. The upper plate **221***a* and the lower plate **222***a*, and the upper plate **221***b* and the lower plate **222***b* are driven at frequencies that vibrate in opposite phases, as in the first embodiment.
(146) Further, by forming the upper plate **221***a* and the lower plate **222***a*, and the upper plate **221***b* and the lower plate **222***b* symmetrically with respect to the central axis (Z-axis) of the ultrasonic vibrator **201**, in the deformation of the diaphragm through the vibration in the Z-axis direction of the ultrasonic vibrator **201**, the vibration between the upper plate **221***a* and the lower plate **222***a* and the vibration between the upper plate **221***b* and the lower plate **222***b* are approximately the same.
(147) When washing the sample nozzle **22**, the sample nozzle **22** is inserted into one of the two washing holes **210***a* and **210***b*. Two sample nozzles **22** can also be washed at the same time.
(148) In addition, in the eighth embodiment, two pairs of diaphragms (two washing holes **210***a* and **210***b*) are used. As in the first embodiment, a pair of diaphragms (one washing hole **210**) may be used.

(149) For example, it is a shape in which the upper plate **221***b* and the lower plate **222***b* are removed and one pair of diaphragms is absent.
(150) Further, when two pairs of diaphragms are installed and one pair of diaphragms is used, one diaphragm may be shaped like a plate (for example, the washing hole **210***b* is closed), or one diaphragm may be shaped like a block (the region between the upper plate **221***b* and the lower plate **222***b* is closed).
(151) Also, in the eighth embodiment, the number of pairs of diaphragms and the installation angle of the main body of the front mass **204** can be changed.
(152) As described above, even in the eighth embodiment, it is possible to radiate ultrasonic waves from a plurality of vibration surfaces, radiate ultrasonic waves in the same phase, and effectively generate cavitation around the sample nozzle **22** through ultrasonic vibrations.
(153) In addition, the present invention is not limited to the above-described embodiments and includes various modifications. For example, the above-described embodiments are specifically described in order to describe the present invention in an easy-to-understand manner and are not necessarily limited to those having all the configurations described.
(154) Also, a part of the configuration of one embodiment can be replaced with a part of the configuration of another embodiment. Also, the configuration of another embodiment can be added to the configuration of one embodiment. Also, a part of the configuration of each embodiment can be deleted, a part of another configuration can be added, and a part of another configuration can be replaced.
REFERENCE SIGNS LIST
(155) **10**: automatic analysis device **11**: reagent container **12**: reagent disk **13**: reaction disk **14**: reagent dispensing mechanism **15**: sample dispensing mechanism **21**: reagent nozzle **22**: sample nozzle **23**: sample container **24**: sample rack **25**: cell **26**: ultrasonic washer **201**: ultrasonic vibrator **202**: washing tank **203**: base **204**: front mass **205**: piezoelectric element **206**: electrode **207**: back mass **208**: volt **209**: flange **210**, **210***a*, **210***b*: washing hole **211**: channel **221**, **221***a*, **221***b*: upper plate **222**, **222***a*, **222***b*: lower plate **401**: resonance frequency f**1** **402**: resonance frequency f**2** **403**: resonance frequency f**3**

Claims

1. An ultrasonic washer comprising: a washing tank for washing a nozzle for sucking a sample or a reagent; and an ultrasonic vibrator in which a piezoelectric element is sandwiched between a front mass and a back mass, wherein diaphragms of an upper plate and a lower plate facing each other are formed at a leading end of the front mass installed at a leading end of the ultrasonic vibrator, the nozzle is ultrasonically washed in a region between the upper plate and the lower plate, and a drive frequency controller to generate a drive frequency of the ultrasonic vibrator is set between the resonance frequency of the upper plate and the resonance frequency of the lower plate.
2. The ultrasonic washer according to claim 1, wherein the ultrasonic vibrator is installed at an angle approximately horizontal and installed so that the angle between ultrasonic irradiation surfaces of the diaphragms and a liquid level of a washing liquid filled in the washing tank are approximately parallel to each other.
3. The ultrasonic washer according to claim 1, wherein the diaphragms of the upper plate and the lower plate are different in length or thickness.
4. The ultrasonic washer according to claim 1, wherein the diaphragms of the upper plate and the lower plate are driven at frequencies that vibrate in opposite phases.
5. The ultrasonic washer according to claim 4, wherein a drive frequency of the ultrasonic vibrator is a resonance frequency at which displacement of the diaphragms of the upper plate and lower pate is maximized.
6. The ultrasonic washer according to claim 1, wherein the upper plate or the lower plate has a

shape that is partially thick.
7. The ultrasonic washer according to claim 1, wherein a difference in frequency between the resonance frequency of the upper plate and the resonance frequency of the lower plate is within 10 kHz.
8. The ultrasonic washer according to claim 4, wherein the upper plate or the lower plate has a maximum amplitude approximately at a position where the nozzle is inserted.
9. The ultrasonic washer according to claim 4, wherein a width of at least one of the diaphragms is wider than a width of a base of the front mass on which the diaphragm is installed.
10. An automatic analysis device characterized by comprising the ultrasonic washer according to claim 1.
11. The automatic analysis device according to claim 10, wherein the ultrasonic washing is performed while changing a height of the nozzle when the nozzle is ultrasonically washed in a region between the upper plate and the lower plate.